

More formally

f is continuous at a
means

$$\lim_{x \rightarrow a} f(x) = f(a)$$

or in terms def. of lim.

$$\forall \varepsilon > 0 \exists \delta > 0 \quad |x - a| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon$$

We have good theorems on how
the algebra of functions, incl. composition,
preserve continuity.

eg. If f, g continuous at a .

then so are.

• $(\alpha f + \beta g)$, for any $\alpha, \beta \in \mathbb{R}$

• $f g$

• $\frac{f}{g}$, provided we avoid
zero in g as before.

For compositions.

Suppose $f, g: \mathbb{R} \rightarrow \mathbb{R}$.



